The Impact of the Pradhan Mantri Jan Dhan Yojana (PMJDY) on Formal and Informal Debt

Context, Motivation, and Objectives

Financial exclusion remains a critical barrier to development in India. Despite sustained economic growth, a significant portion of the population, especially those in rural areas, has historically lacked access to formal savings and credit instruments, forcing them to rely on high-cost informal lenders, such as moneylenders. In 2014, the Government of India launched the Pradhan Mantri Jan Dhan Yojana (PMJDY), an intervention designed to universalize banking access. The theoretical motivation for this study is to determine whether access to banking infrastructure translates into actual credit usage.

The objective of this paper is to estimate the causal impact of PMJDY implementation intensity on household borrowing behavior. Specifically, the outcome variables are the district-level aggregate amounts of formal debt (loans from banks, cooperatives, and government agencies) and informal debt (loans from moneylenders, landlords, and relatives). Crucially, these variables capture the flow of new borrowing rather than the total stock of outstanding debt. I filter the AIDIS transaction records to include only loans made in the calendar year preceding the survey (2011 for the 70th Round and 2018 for the 77th Round). This ensures the analysis reflects active credit market behavior during the PMJDY implementation period. The treatment variable is the state-level PMJDY intensity, defined as the ratio of beneficiaries to the population.

Data Description & Summary Statistics

This study utilizes district-level aggregated data from the All India Debt and Investment Survey (AIDIS) conducted by the National Sample Survey Office (NSSO). I construct a panel dataset using two rounds: the 70th Round (2013), serving as the pre-treatment baseline, and the 77th Round (2019), representing the post-treatment period five years into the PMJDY rollout. Historical data from the 2003 AIDIS round are also employed for testing assumptions. This was combined with state-level measures of PMJDY intensity using parliamentary records on the number of Jan Dhan beneficiaries in 2014 and normalized using population counts from the 2011 Census. To classify districts into treatment categories, I divide them as either belonging to high or low intensity states, using a median intensity threshold of 22 percent beneficiaries per capita.

The summary statistics reveal three noteworthy points:

First, there is a clear trend toward the formalization of credit. As shown in Table 1, the average formal borrowing per district increased by 64.2% from 2012 and 2018 (from ₹45.7 lakh to ₹75.1 lakh). In contrast, informal borrowing grew at a much slower rate of 25.6%. This suggests that the expansion of formal credit was a nationwide phenomenon, independent of specific state-level interventions. Increased sources of formal credit accessible to households were also evident in the rise in the number of formal credit sources available to households, as seen in the metadata of AIDIS rounds over the three years.

Second, there is significant baseline heterogeneity between high- and low-intensity states. As detailed in Table 2, districts in low-intensity states initially had double the formal debt of those in high-intensity states in 2013 (₹66.6 lakh vs. ₹33.8 lakh) even when the total number of districts in high-intensity states was more. This confirms that the PMJDY rollout was endogenous to financial development: the government targeted the program most aggressively in states that were historically under-banked.

Third, the data highlights a rural-urban divide. Table 3 shows that in the pre-treatment year (2013), urban districts held nearly three times the amount of formal debt as rural districts (₹95.7 lakh vs. ₹33.0 lakh).

Causal Model

The causal structure of this study is that the intensity of PMJDY implementation has a direct effect on financial access. An increase in financial access (via bank account ownership) is hypothesized to increase formal credit and decrease informal credit, where households switch from high-interest informal loans to lower-interest formal loans.

However, unobserved state-level characteristics and historical banking infrastructure simultaneously influence both the intensity of PMJDY rollout and the credit market development. A naive comparison of high vs. low intensity states is therefore confounded, which likely creates a downward bias (since poorer states got more PMJDY but also have lower credit growth). To identify the causal effect, I thus look at those who have the most to gain from the policy: the rural population.

Methodology

For a canonical DiD to provide a correct estimate, the high-intensity states (treated) and low-intensity states (control) must exhibit similar growth trends prior to the policy's implementation. Essentially, if the policy never happened, their lines on a graph should move in parallel. However, data from 2003 to 2013 revealed that this assumption did not hold. Thus,

canonical DiD would fail to give an unbiased estimand of interest i.e. Average Treatment Effect on the Treated (ATT).

The Triple Difference methodology addresses this by leveraging the rural-urban divide highlighted in Table 3. Since PMJDY is primarily targeted at the unbanked rural population, urban districts within the same high-intensity states can serve as an internal control group. By subtracting the "urban trend" from the "rural trend" within districts in high-intensity states and comparing that to the same difference in districts in low-intensity states, I isolate the rural-specific impact of PMJDY while netting out all state-level economic shocks that affect both sectors equally.

The primary regression equation is estimated as follows:

$$
\begin{aligned}
Y_{id} = {} & \beta_1 \left(\text{HighIntensity} \times \text{Rural} \times \text{Post}\right) \\
& + \beta_2 \left(\text{HighIntensity} \times \text{Rural}\right) + \beta_3 \left(\text{HighIntensity} \times \text{Post}\right) + \beta_4 \left(\text{Rural} \times \text{Post}\right) \\
& + \beta_5 \, \text{HighIntensity} + \beta_6 \, \text{Rural} + \beta_7 \, \text{Post} \\
& + \theta_t \, \text{YearFixedEffects} + \alpha_s \, \text{StateFixedEffects} + \varepsilon_{id}
\end{aligned}
$$

The coefficient of interest is β1, which captures the ATT: the differential change that PMJDY intervention caused on average (formal/informal) debt for rural districts in high-intensity states relative to urban districts, over and above the counterfactual rural-urban trend established by low-intensity states.

Identifying Assumptions

The causal validity of the DDD estimator relies on the assumption of parallel trends in differences. DDD requires that, in the absence of the policy, the gap between rural and urban credit growth would have evolved similarly across high and low-intensity states.

This assumption is weaker and more plausible than standard DiD. While high-intensity states and low-intensity states have different baseline debt levels and growth rates, it is reasonable to assume that the internal structural relationship between their rural and urban economies evolves consistently. For example, if a national agricultural shock affects rural areas, it should affect rural areas in both high and low-intensity states similarly relative to their urban centers.

I empirically validate this assumption using a pre-trend test on data from 2003 to 2013. Running the full DDD specification on this pre-period yields a statistically insignificant coefficient for the triple interaction term. This null result confirms that prior to PMJDY, the rural-urban credit gap was not changing differentially between state groups. Consequently, any significant

triple-difference observed after 2014 can be attributed to the policy intervention rather than pre-existing differential trends.

Furthermore, I assume no spillovers: the treatment of rural areas does not impact the credit access of urban areas (SUTVA).

Results and Interpretation

Formal Credit Results: The estimation of the impact of PMJDY on formal borrowing reveals a coefficient of 76.61 lakh with a p-value of 0.121. Despite the massive increase in bank account ownership documented in administrative data, the estimator suggests no statistically significant increase in the aggregate amount of formal debt in rural areas of high-intensity states compared to the counterfactual. While the coefficient is positive, the large standard errors indicate substantial heterogeneity, and we cannot reject the null hypothesis that the policy had no impact on formal credit volumes at the district level.

Informal Credit Results: Similarly, for informal debt, the DDD coefficient is -5.60 lakh (p-value: 0.354). I find no significant evidence that PMJDY intensity caused a reduction in informal borrowing amounts. The hypothesized substitution effect, where households switch from moneylenders to banks upon receiving an account, is not visible in the aggregate district-level data for the 2013-2019 period. The coefficient is negative, but it is statistically indistinguishable from zero.

The null results can be interpreted through two lenses.

First, access vs. usage: PMJDY succeeded as a supply-side intervention by providing the infrastructure of banking (accounts and debit cards). However, credit underwriting relies on demand-side factors, such as collateral, credit history, and regular income documentation. Newly banked rural households may possess an account but still lack the creditworthiness required for formal loans, leaving their borrowing capacity unchanged.

Second, possible long-term effects: the 2019 survey occurred only five years after the program's launch. In banking, establishing an active credit account often takes longer as trust and transaction history are built up. It is plausible that the credit effects of PMJDY will only materialize over a longer time horizon.

Robustness Checks

To ensure these results were not driven by model misspecification, two placebo tests were performed for robustness, and a pre-trend validation was mentioned earlier.

Placebo Treatment Group: I restricted the sample to only low-intensity (control) states and randomly assigned half of them to a fake high-intensity group. Re-running the DDD model on this sample yielded a statistically insignificant coefficient ($p = 0.176$). This result confirms the validity of the estimator: the model correctly identifies no effect when applied to a sample where no differential treatment existed.

Placebo Outcome Test: I replaced the dependent variable with average household size, a demographic variable that should theoretically be unaffected by banking policy in the short run. The estimated DDD coefficient for this outcome was -0.09 with a p-value of 0.615. This null effect confirms that the model does not mechanically generate significant results where none should exist.

Pre-trend Validation: As detailed in the methodology, the full DDD model was run on 2003-2013 data. The coefficient on the triple interaction term was statistically insignificant for both formal and informal credit. This null result in the pre-period confirms that the estimator is not picking up spurious historical trends, validating the research design.

Conclusion

Using a Triple Difference methodology, I find that while PMJDY dramatically increased financial coverage (bank account ownership), it did not lead to a statistically significant causal increase in aggregate formal credit volume or a decrease in informal credit volume between 2013 and 2019. The null results highlight a critical issue with government initiatives at large: access to infrastructure does not automatically translate into usage. Providing a bank account is a necessary condition for formal credit, but it is not sufficient. Structural barriers such as the lack of collateral, irregular rural incomes, and the informational advantages of local informal lenders appear to remain dominant forces in India's credit landscape.

A key limitation of this study is the reliance on district-level aggregate loan amounts. It is possible that PMJDY increased the number of small loans without significantly moving the total volume, which might be dominated by wealthier households. Furthermore, the binary classification of high vs low intensity based on a 2014 snapshot captures the initial implementation but may mask improvements in lower-intensity states over the subsequent five years. Additionally, as noted in the results, the large standard errors suggest substantial heterogeneity, meaning the policy may have worked in specific sub-regions not visible at the aggregate level. By focusing on fresh borrowings in 2012 and 2018, this study analyzes credit

flow rather than stock. While this accurately measures immediate policy impact, it does not capture changes in the repayment or persistence of older legacy debt.

Future research should prioritize the use of household-level microdata to disentangle the effects on marginal borrowers. Specifically, studies should examine the probability of having a loan rather than just volumes to detect if the policy democratized access for the poor. Moreover, longer-term studies are needed to test the "lagged effects" hypothesis. Finally, qualitative research is essential to understand the supply-side friction: are bankers reluctant to lend to new Jan Dhan account holders, or are account holders simply not demanding formal credit?

Tables

Debt amounts represent the aggregate value of new loans taken during the reference year (2012 for 2013 round, 2018 for 2019 round).

Table 1: Average District-Level Debt (2012 vs. 2018)

Year	Average Formal Debt (₹ Lakhs)	Average Informal Debt (₹ Lakhs)	Total Debt (₹Lakhs)	Number of Districts
2013	45.75	16.62	62.37	632
2019	75.14	20.87	96.02	681
Growth(%)	64.2	25.6	53.9	

Table 2: Baseline Imbalance by Treatment Intensity (2013)

State Category	Mean Formal Debt (₹)	Mean Informal Debt (₹)	Total Debt (₹)
Low Intensity	66.58	22.35	88.93
High Intensity	33.78	13.42	47.2
Difference	-32.8	-8.93	-41.73

Table 3: The Rural-Urban Credit Gap (Baseline 2013)

Sector	Mean Formal Debt (₹)	Mean Informal Debt (₹)
Urban	95.73	23.86
Rural	33.06	14.78
Gap (Rural - Urban)	-62.67	-9.08

Table 4: Main DDD Regression Results (2013-2019)

Variable	Formal Debt (₹ Lakhs)	Informal Debt (₹ Lakhs)
Triple Diff (Intensity*Post *Rural)	76.61	-5.6
P-value	0.121	0.354
Observations	1253	1253

Table 5: Placebo Tests (Robustness)

Test Type	Outcome Variable	DDD Coefficient	P-value
Pre-Trend (2003-2013)	Formal Debt	8.38	0.696
Fake Outcome	Avg. Household Size	-0.09	0.615

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